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Experimental Modal Analysis of a Light Train Wheel

ADVANCED DYNAMICS OF MECHANICAL SYSTEMS
LAB ACTIVITY REPORT

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1 | Structure Analyzed

The system analyzed is a wheel of a light train vehicle. The aim of the modal experimental analysis is to characterize the wheel, in order to better understand its dynamic behavior and use the data collected for noise studies, in particular on the rolling and on the squeal noise issues. The wheel considered is a **resilient wheel**: that means that it is not forged from a unique piece of steel, but the rim is separate from the wheel web by a rubber pad, that has a damping function.

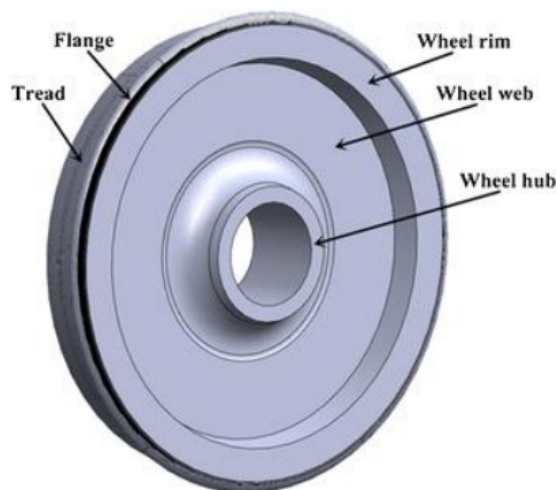


Figure 1.1: Resilient Wheel

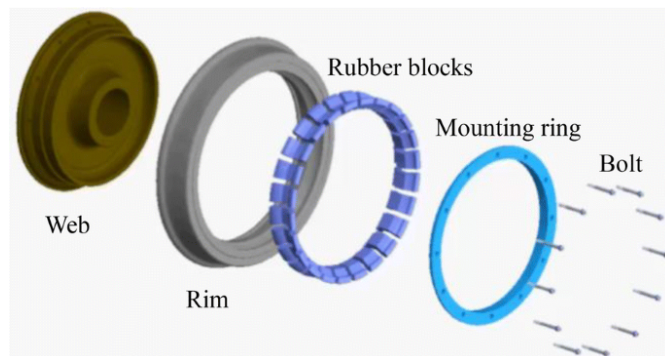


Figure 1.2: Resilient wheel components

1.1. Experimental set-up

The experimental data are collected in laboratory, considering a **free-free** constraints set-up. The wheel is suspended through four springs, with an elastic constant selected carefully, in order to not introduce a resonance in the frequency range of interest. The impulse is applied with a dynamometric impact hammer, and the vibration data are collected thanks to 12 accelerometers placed on the resilient wheel. The measuring grid covers only half of the wheel, as shown in Figure 1.4, taking advantage of the symmetry of the body.



Figure 1.3: Experimental set-up



Figure 1.4: Measuring grid

1.2. Experimental FRF

In Figure 1.5 are reported the experimental Frequency Response Function collected in the experimental campaign, in terms of amplitude, phase and coherence function. The first, second and fourth resonance peaks are clearly visible and identifiable, while the third present a shift of a few hertz in the peaks of some FRF. This behavior is due to two different vibration modes that have resonance in a narrow frequency range. Moreover, it is possible to note some drops in the coherence function, for example in correspondence to the accelerometers placed near to a vibration node.

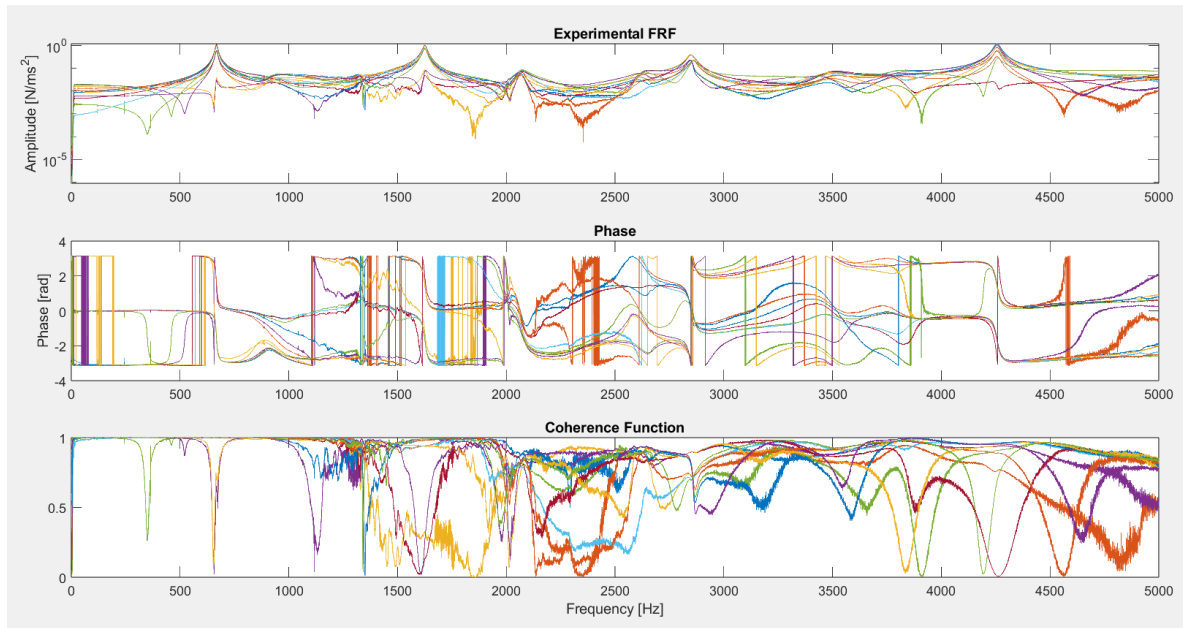


Figure 1.5: Experimental FRF

2 | Experimental Identification

2.1. Definition of the Experimental Matrix

We define the Experimental Matrix $[G_{exp}]$ as a matrix that collects by columns the twelve Frequency Response Functions given in output by the accelerometers.

2.2. Definition of the Initial Guesses

In order to proceed with the modal identification analysis, we have to provide to the minimization algorithm a vector, $\underline{X}_0$ of initial guesses. The vector has a length of $3 \cdot 12 + 2 = 38$ elements. Since some of the FRF given are colocated FRFs, we decide to estimate the guesses for ω_i and ξ_i using only the first FRF, in which all the peaks are well defined. We expect to provide a good guess for the first, the second, and the fourth natural frequency, while the third mode will be affected by errors due to the superposition of modes with slightly different natural frequencies.

- **Guess for ω_i :** obtained from the simplified procedure applied on the first FRF, using the Matlab function *findpeaks*;
- **Guess for ξ_i :** obtained from the simplified procedure on the first FRF, looking at the slope of the phase in correspondence to ω_i : $\xi_i = -1/\omega_i \cdot (1/d(phase))$.
- **Guess for A_{jk} :** obtained from the simplified procedure as: $A_{jk} = -imag[G_{exp}(\omega_i) \cdot (2\omega_i^2 \xi_i)]$;
- **Guess for the residuals:** put to zero both for R^L and R^H .

As a check for the guesses estimated from the data, it's possible to plot a first version of the numerical curve, based only on the values inside $\underline{X}_0$. As an example, in Figure 2.1 the result for the first peak of the first FRF is reported.

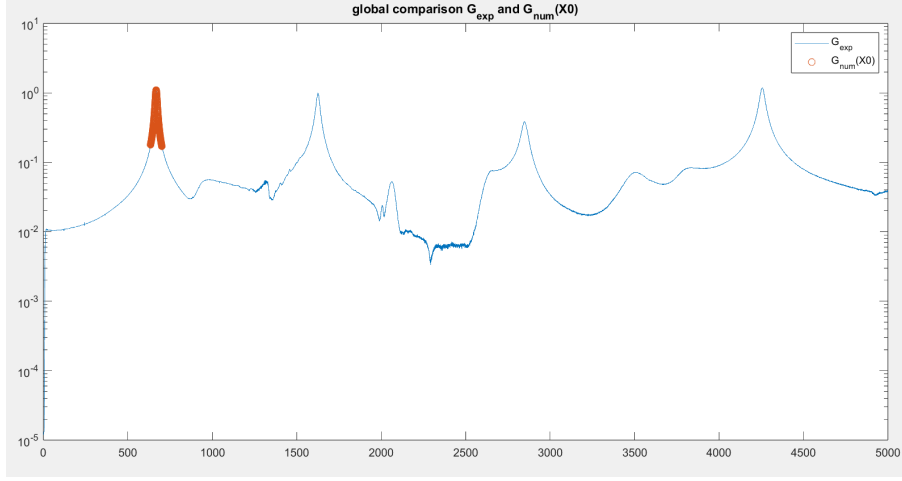


Figure 2.1: Comparison $G_{exp}(:, 1)$ and $\underline{X}_0$

2.3. Error Function and Minimization

In order to define the estimated numerical curve G_{num} that fits the experimental data stored in G_{exp} we rely on a least squares minimization method, using the built-in Matlab function *lsqnonlin*. This function provides the numerically computed values of ω_i , ξ_i , A_{jk} , and residuals that best fit the data around each resonance peak. In the table below the results concerning the natural frequencies and the non-dimensional damping coefficients are shown:

	COMPUTED VALUES
ω_1	667.32 Hz
ω_2	1625.29 Hz
ω_3	2847.92 Hz
ω_4	4256.28 Hz
ξ_1	0.0080
ξ_2	0.0060
ξ_3	N.D.
ξ_4	0.0036

Table 2.1: Values for ω_i and ξ_i

2.4. Graphical Outputs

In the following figures, some of the graphical results obtained in Matlab are reported. We choose to report only some of the 12 FRF analyzed, as an example. The identification

procedure didn't work well on the third peak, as expected; the results are anyway showed in Subsection 2.4.3.

2.4.1. First peak

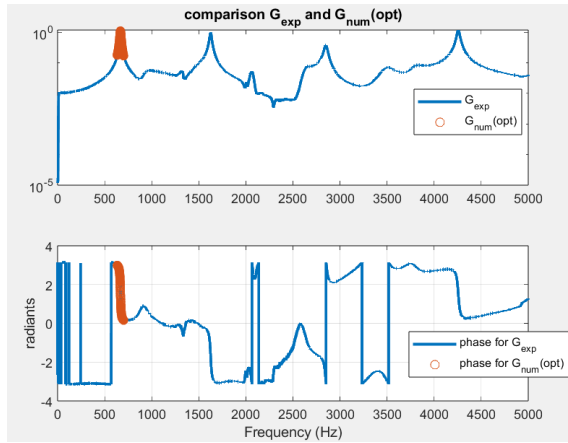


Figure 2.2: 1 FRF

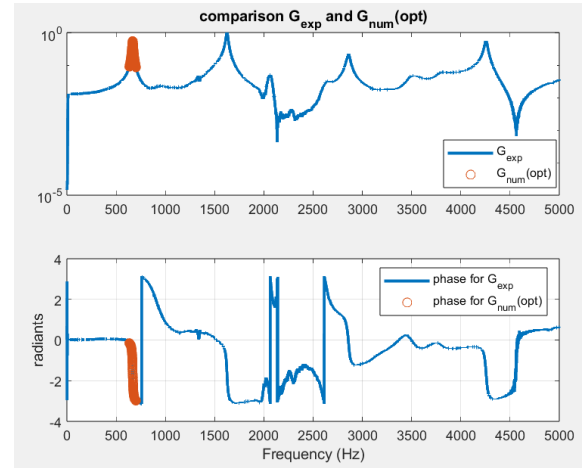


Figure 2.3: 2 FRF

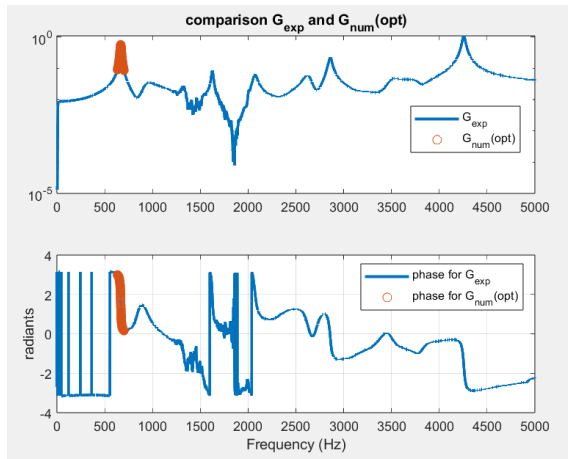


Figure 2.4: 3 FRF

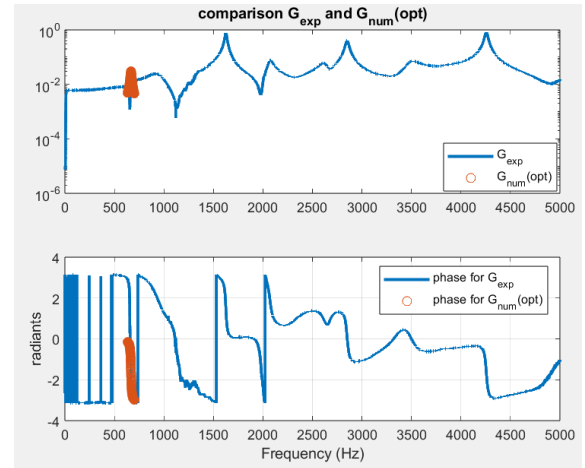


Figure 2.5: 4 FRF

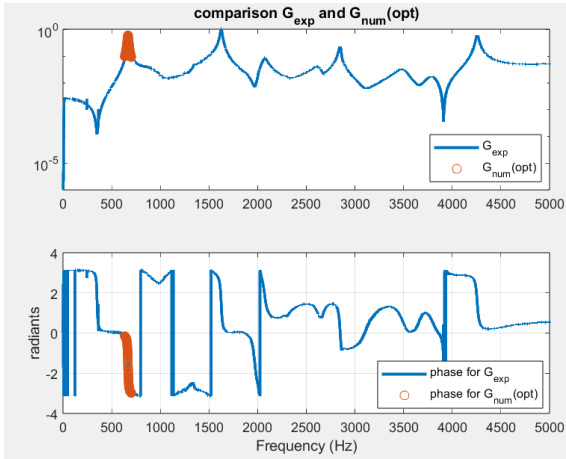


Figure 2.6: 5 FRF

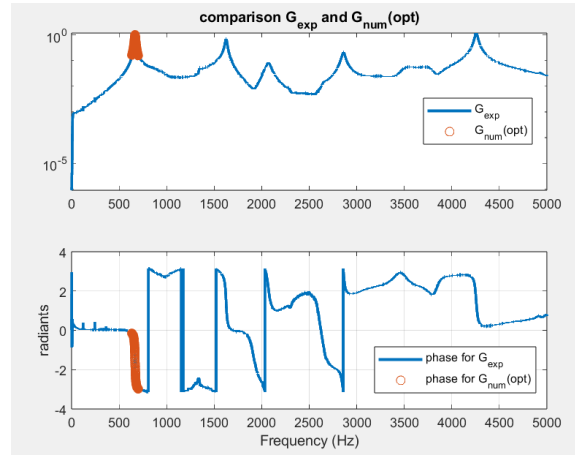


Figure 2.7: 6 FRF

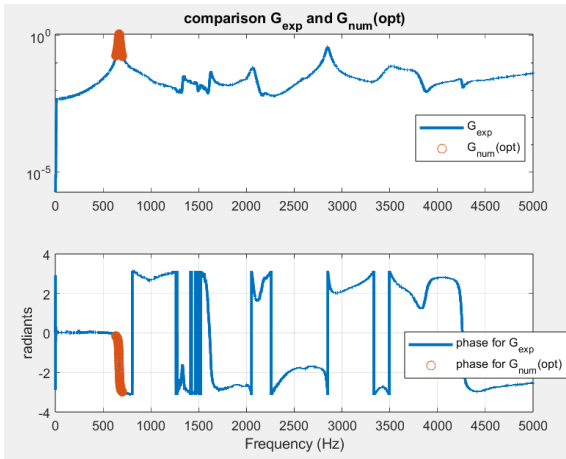


Figure 2.8: 7 FRF

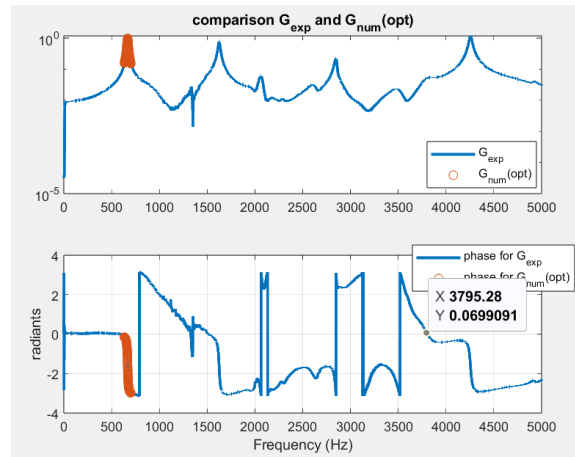


Figure 2.9: 8 FRF

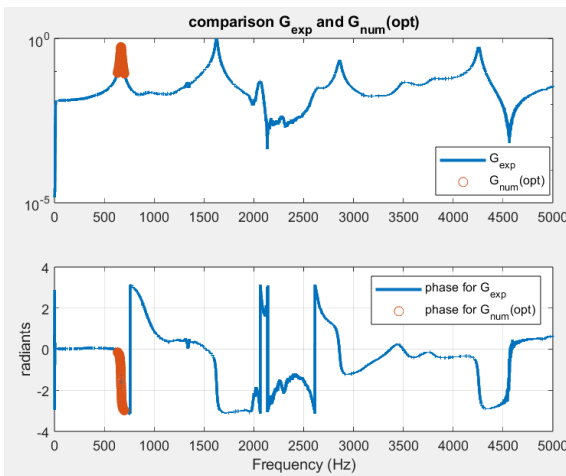


Figure 2.10: 9 FRF

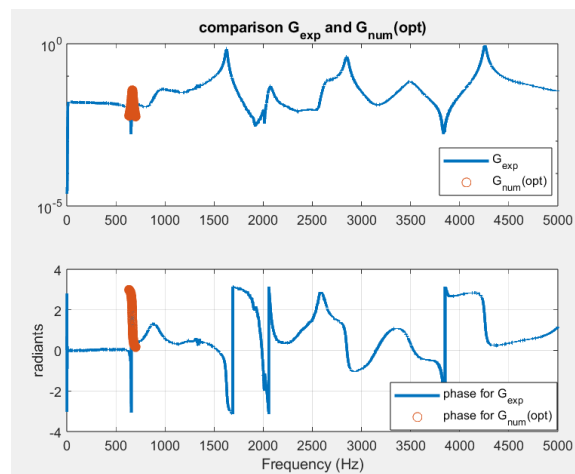


Figure 2.11: 10 FRF

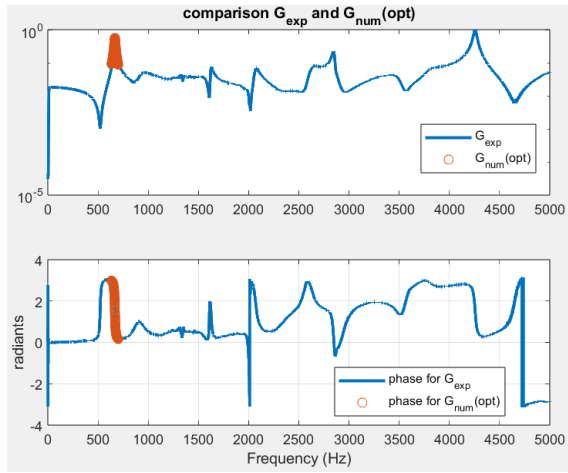


Figure 2.12: 11 FRF

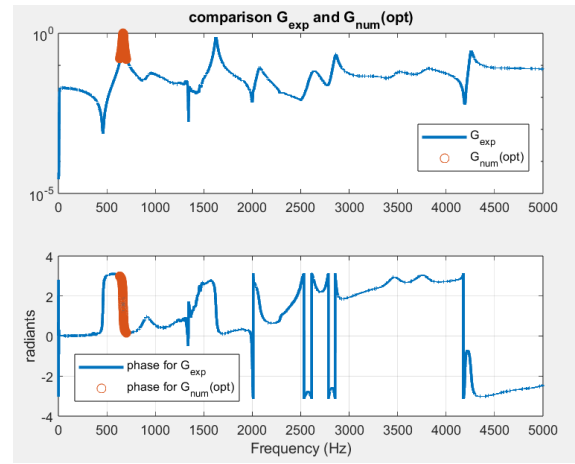


Figure 2.13: 12 FRF

2.4.2. Second peak

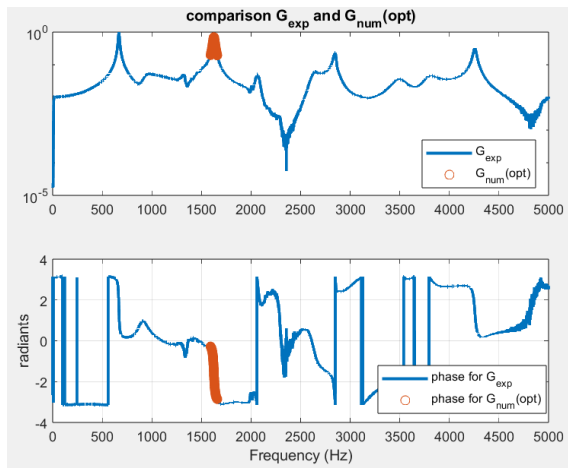


Figure 2.14: 2 FRF

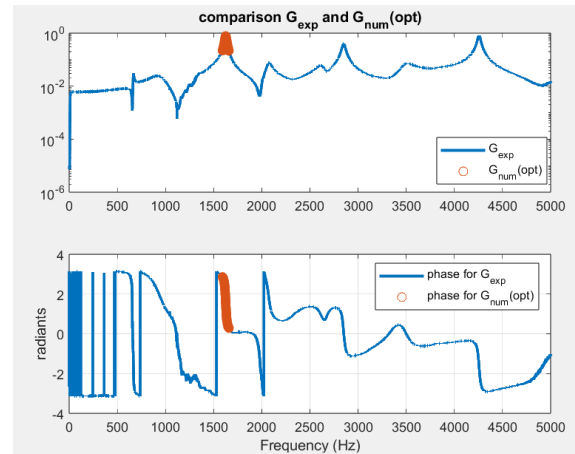


Figure 2.15: 4 FRF

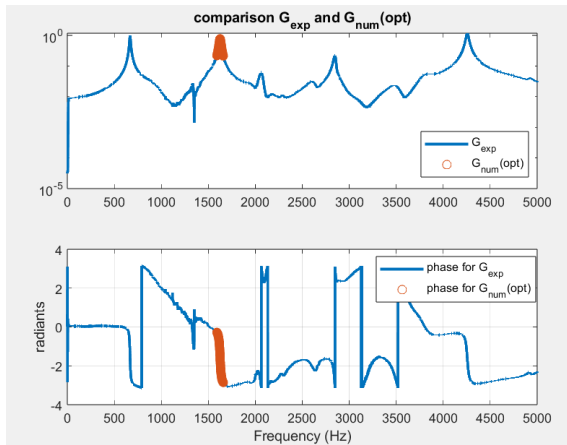


Figure 2.16: 8 FRF

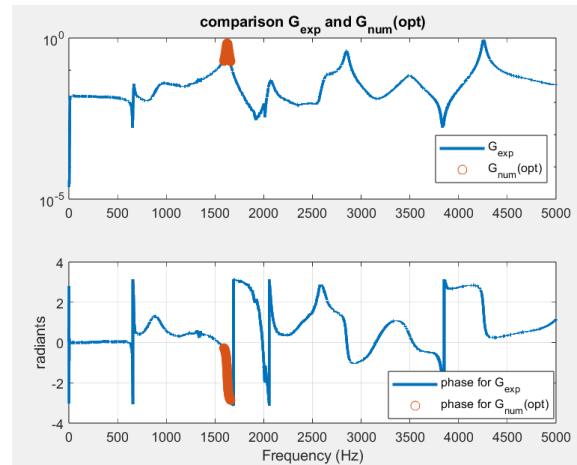


Figure 2.17: 11 FRF

2.4.3. Third peak

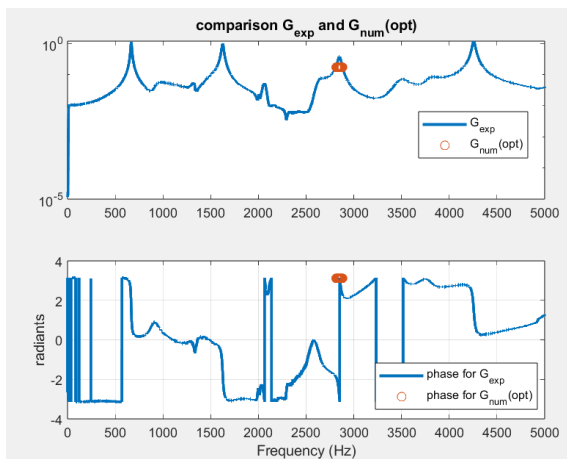


Figure 2.18: 1 FRF

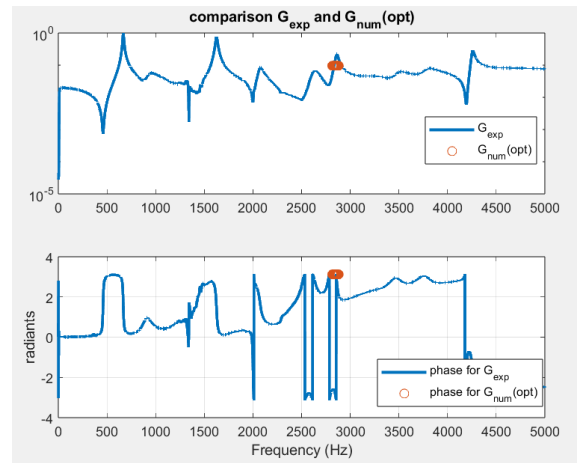


Figure 2.19: 12 FRF

2.4.4. Fourth peak

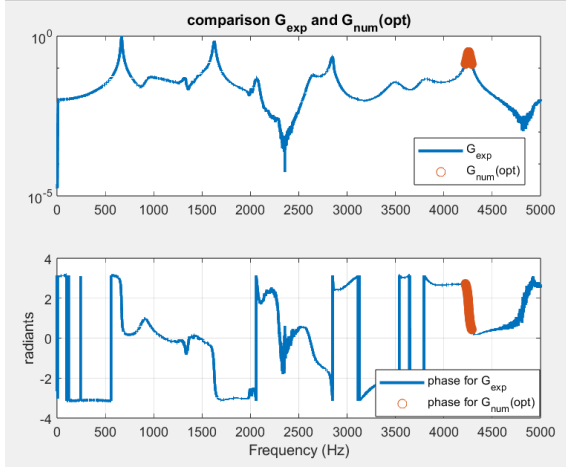


Figure 2.20: 2 FRF

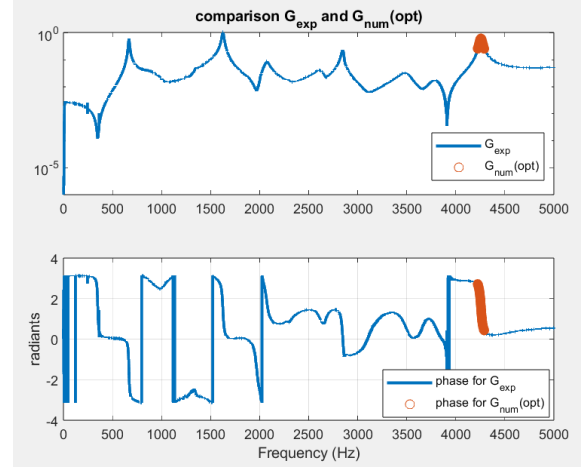


Figure 2.21: 5 FRF

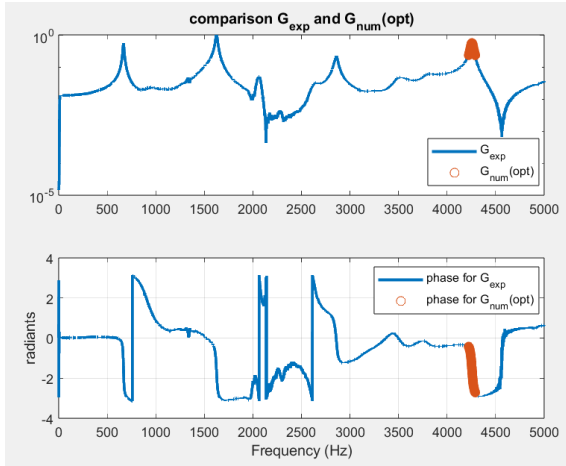


Figure 2.22: 9 FRF

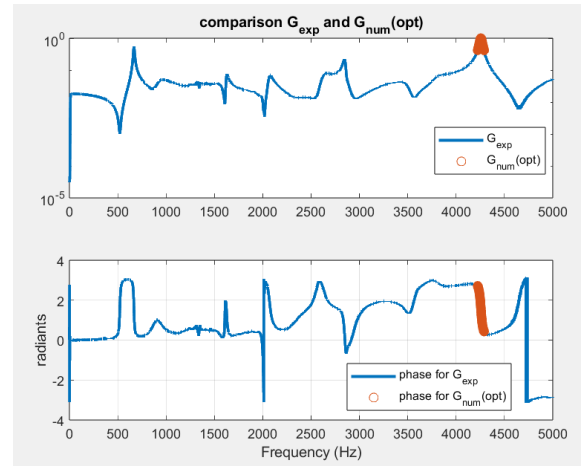


Figure 2.23: 11 FRF

2.5. Identified Mode Shapes

Starting from the data in output from the solver, *lsqnonlin* it is possible to reconstruct the mode shapes of the wheel. With a proper scaling factor the A_{jk} numerically computed are plotted in a **polar plot** in correspondence to the position of the sensors. Due to the **symmetry** of the resilient wheel, the points found are then mirrored on the other half. Since the sensor in correspondence of the co-located FRF was missing, the plots of the modes are not closed lines. As a first approach, the modes are plotted as simple points, interpolated by a straight line. In a more refined version, it is possible to see the

same modes interpolated with a **cubic spline**. The points in correspondence with the accelerometers are highlighted. As the natural frequency increases, we see the number of **nodal diameters** increasing. In particular, since the mode identified in correspondence of the first peak has two nodal diameters, it is possible to conclude that is the second mode of the wheel. In the following pictures also the **fourth mode** is reported, even if it has been found with errors in the minimization algorithm: we don't expect this shape to be exactly the real one.

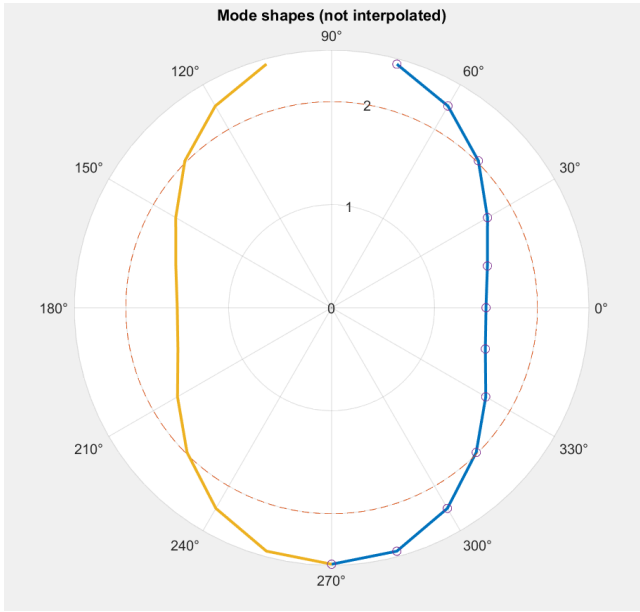


Figure 2.24: Second mode (not int.)

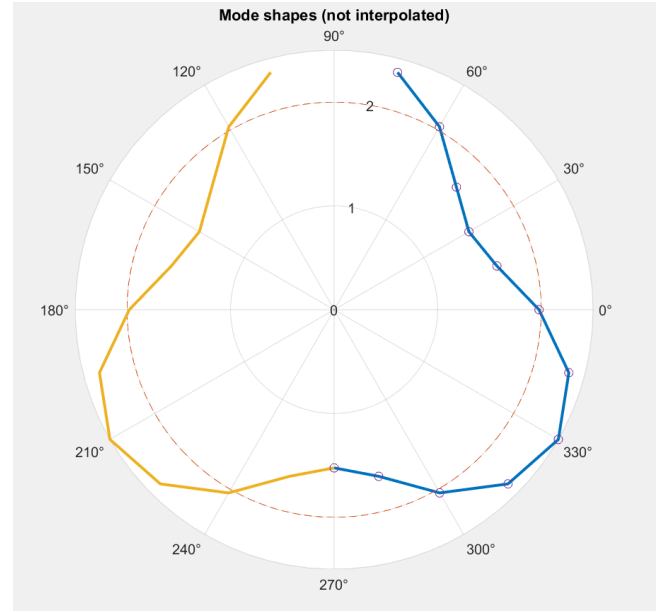


Figure 2.25: Third mode (not int.)

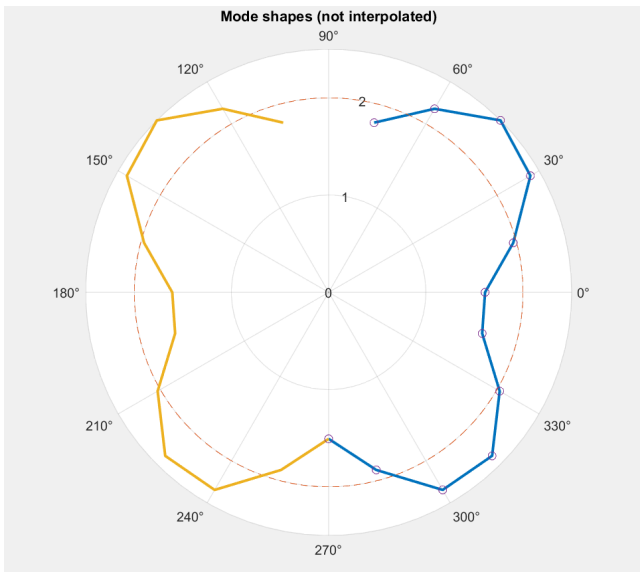


Figure 2.26: Fourth mode (not int.)

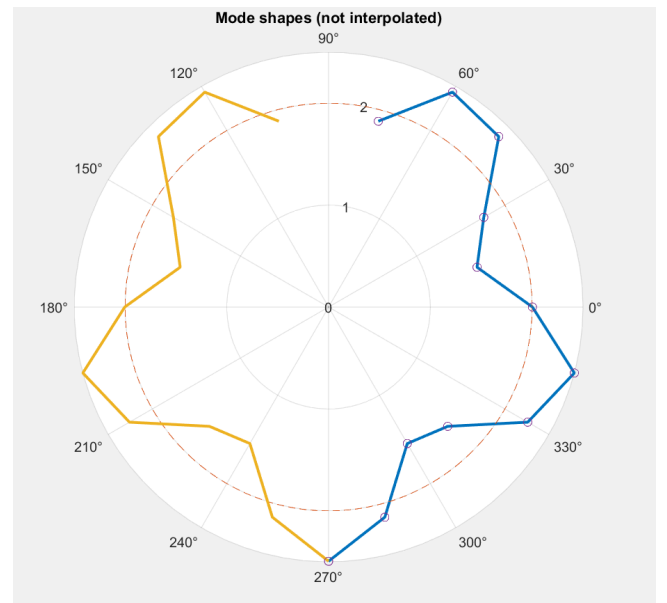


Figure 2.27: Fifth mode (not int.)

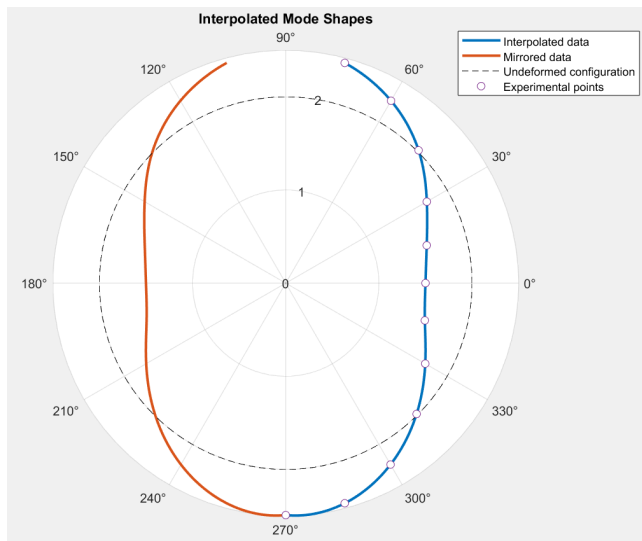


Figure 2.28: Second mode

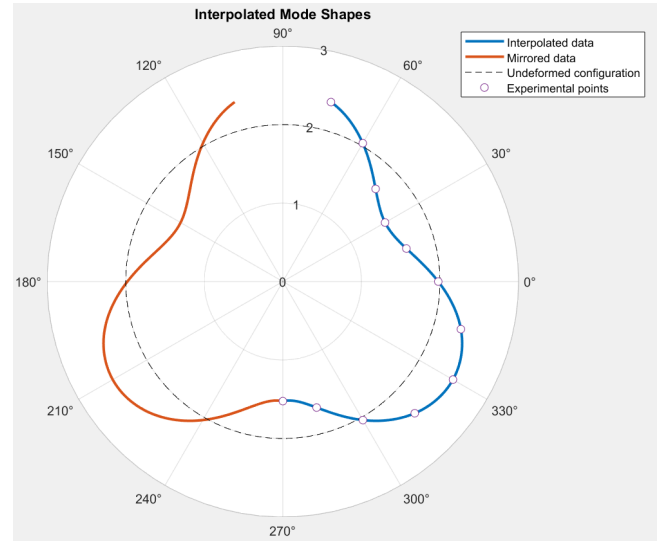


Figure 2.29: Third mode

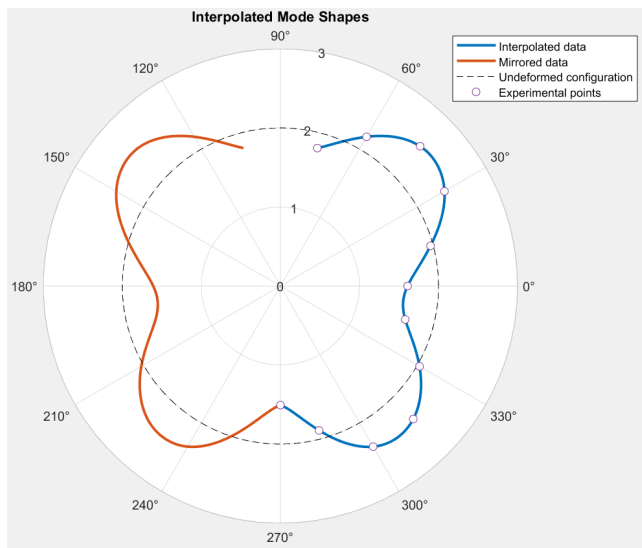


Figure 2.30: Fourth mode

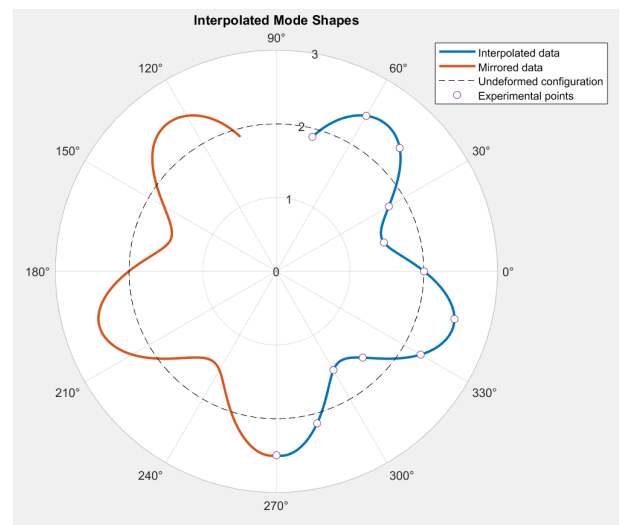


Figure 2.31: Fifth mode

